Assignment 7: Week 0 Career Challenge

Weekly Goal

Completed and refined the Mercer General Emergency Triage clinical data pipeline while maintaining a sustainable 20-hour workweek.

Weekly Focus Areas

1. Clinical Data Cleaning
2. Physiological data analysis
3. Visualisation development
4. Predictive triage logic
5. At-risk rule pseudocode
6. Workflow consistency and management

Total Protected Work Hours:

Approx. 20 Hours

Week 0 Project Timeline

Day	**Focus Area**	**Main Objective**	Hours
Wednesday	Gender Data Cleaning	Environment setup and gender data normalisation	4
Thursday	Group Work - Respiratory Rate Cleaning	Group work: Respiratory Rate (RR) cleaning and data quality assurance	2
Friday	Clinical Analysis and Visualisation	Emergency Department triage data cleaning and visualisation	4
Saturday	Mean Arterial Pressure (MAP) Analysis	Mean arterial pressure (MAP) analysis and interpretation	2.5
Sunday	Oxygen Saturation (SpO2) Analysis and Development	Analyse and interpret Oxygen Saturation (SpO2)	2.5
Monday	Triage Pseudocode	Developed an at-risk rule-based pseudocode for the emergency department triage	4

Tuesday	Weekly Audit and Finalisation		1.5

Week 0 total hours:

20.5 hours

Week 0 Work Schedule

Wednesday - May 20th 2026

<u>Time block</u>

8:00 AM - 9:00 AM : Welcome to Carisurg Meeting
9:30 AM - 10:30 AM : Week 0 Challenge Overview
11:00 AM - 12:00 PM : Tutorial 1 - Python for basic data exploration and cleaning
3:00 PM - 3:30 PM : Daily standup

<u>Main objective</u>

Environment setup and gender data normalisation

<u>Tasks</u>

1. Mounted Google Drive and configure Colab workspace
2. Imported essential python libraries: pandas, numpy and matplotlib
3. Reviewed and validated the raw dataset structure
4. Standardised inconsistent gender values
5. Applied a mapping logic to convert mixed numeric and categoric values into clean labels (“F” and “M”)
6. Replaced inconsistent variables.

<u>Focus Structure</u>

- 4 pomodoro cycles : 1 hour each cycle, 15 mins short break, 30 mins long break

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Thursday - May 21st 2026

<u>Time block</u>

8:00 AM - 9:00 AM : Guest Speaker: Dr. Loren De Freitas (What a Caribbean ED is like)

9:30 AM - 10:30 AM : Tutorial 2 - Python for Advanced Data Cleaning.
3:00 PM - 3:30 PM : Daily Standup

Main objective

Group work: Respiratory Rate (RR) cleaning and data quality assurance

Tasks

1. Verified environment and python version compatibility
2. Review dataset structure and confirmed pipeline readiness for processing
3. Performed numeric constraints for RR to eliminate invalid or non-numeric entires
4. Implemented data validation checks
5. Generated pre and post cleaning distribution plots to visualise data quality improvements
6. Standardised inconsistencies with clinical records.

Focus Structure

- 2 hours straight runthrough

Friday - May 22nd 2026

Time block

8:00 AM - 9:00 AM : Tutorial 3 - Basic Data Visualisation with Matplotlib
11:00 AM - 11:30 AM : Daily Standup
1:00 PM - 2:00 PM : Guest Speaker: Dr. Yom Alemante (Innovating Emergency Medicine in High and Low Resource Settings)

Main objective

Emergency Department triage data cleaning and visualisation

Tasks

1. Standardised and cleaned clinical variables across the dataset.
2. Corrected and validated gender values into binary format.
3. Converted Glasgow Coma Scale (GCS) values into numeric format and filtered values outside of clinical range (3-15)
4. Filtered systolic (SBP) and diastolic (DBP) blood pressure values into clinical ranges
5. Standardised pulse, RR and temperature variables.
6. Converted mixed temperature values (Celsius and Fahrenheit) into a Celsius scale.
7. Performed data analysis to identify abnormal readings.
8. Explored the relationship between SBP and GCS.

    Visualisation Outputs

i. SBP distribution
ii. SBP vs GCS Scatter Plot
iii. SBP by GCS Boxplot

Focus Structure

- 4 pomodoro cycles : 1 hour each cycle, 15 mins short break, 30 mins long break

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Saturday - May 23rd 2026

Time block

7:00 AM - 9:30 AM

Main objective

Mean arterial pressure (MAP) analysis and interpretation

Tasks

1. Define MAP and why it is important in clinical triage
2. Identify the normal and abnormal ranges

Focus Structure

- 25 minutes - Idea Collection
- 30 minutes - Reading and Note Taking
- 30 minutes - Draft Creation
- 30 minutes - Finalising Report
- 20 minutes - Initial Submission Feedback Review
- 15 minutes - Final Adjustment and Resubmission

Sunday - May 24th 2026

Time block

5:00 AM - 7:30 AM

Main objective

Analyse and interpret Oxygen Saturation (SpO2)

Tasks

1. Define SpO2
2. Identify factors that influence oxygen saturation

3. Identify how SpO2 can be an early warning indicator for respiratory distress in triage
4. Apply SpO2 thresholds and classify patient risk levels

Focus Structure

- 25 minutes - Idea Collection
- 30 minutes - Reading and Note Taking
- 40 minutes - Draft Creation
- 35 minutes - Finalising Report
- 15 minutes - Final Adjustment and Submission

Monday - May 25th 2026

Time block

8:00 AM - 8:30 AM : Daily Standup
10:30 AM - 11:30 : Career Challenge - Time Management for Medtech Students

Main objective

Developed an at-risk rule-based pseudocode for the emergency department triage

Tasks

1. Define key physiological inputs
2. Apply clinically accepted at-risk thresholds for each vital sign
3. Build a IF/THEN logic to evaluate patient severity
4. Covert temperature
5. Design triage classification (Critical, High priority, Standard)

Focus Structure

- 4 pomodoro cycles : 1 hour each cycle, 15 mins short break, 30 mins long break

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Tuesday - May 26th 2026

Time block

8:00 AM - 8:30: Daily Stand Up
Flexible / Review Session

Main objective

Reviewing weekly performance and optimise time management

Tasks

1. Review progress from Week 0
2. Assess efficiency
3. Reflect on cognitive load across the different stages of Week 0
4. Use findings to improve future scheduling

Focus Structure

- 1.5 hours straight runthrough

Week 0 Reflection on Workflow

Overall, this week 0 schedule has helped me follow a clear clinical data pipeline, which has improved my understanding of how each stage connects. Applying at least 20-hours a week kept the workload manageable and well-balanced. I noticed my consistency dropped on certain days which affected my focus. From this, I noticed that I work best in uninterrupted blocks and need to fix my schedule going forward to improve my efficiency.